

Stargazing

Astronomy is the branch of science that is dedicated to studying stars, planets, and all the celestial bodies that surround Earth. It seeks to explain where we came from and the beginning of the Universe itself.

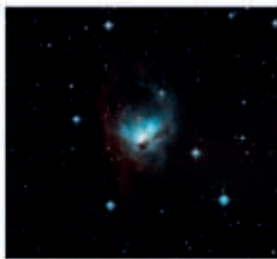
LOOKING AT THE SKY

Binoculars are a great way to start looking at the night sky, because they reveal up to ten times as much detail as the naked eye and are easy to use. Telescopes provide even greater detail.

MERCURY, VENUS, MARS, JUPITER, AND SATURN CAN BE SEEN WITH THE NAKED EYE



NAKED-EYE VIEW OF THE ORION NEBULA



BINOCULAR VIEW OF THE ORION NEBULA



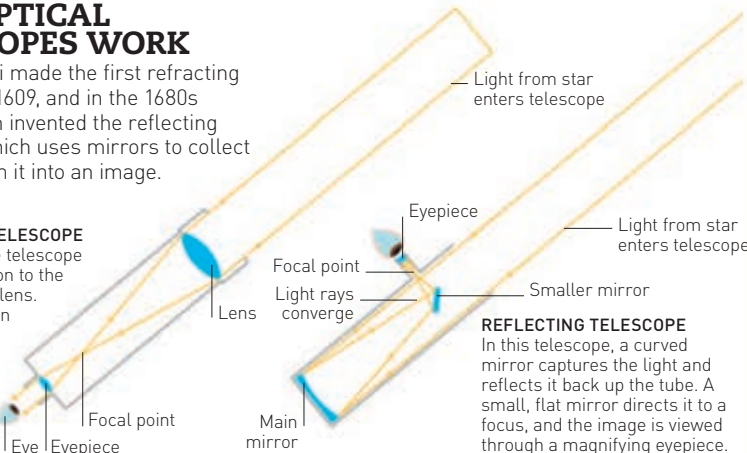
TELESCOPE VIEW OF THE ORION NEBULA

HOW OPTICAL TELESCOPES WORK

Galileo Galilei made the first refracting telescope in 1609, and in the 1680s Isaac Newton invented the reflecting telescope, which uses mirrors to collect light and form it into an image.

REFRACTING TELESCOPE

Light enters the telescope and is focused on to the focal point by a lens. An eyepiece then magnifies the focused light into an image the right size for your eye.



REFLECTING TELESCOPE

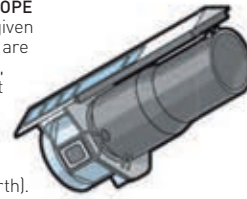
In this telescope, a curved mirror captures the light and reflects it back up the tube. A small, flat mirror directs it to a focus, and the image is viewed through a magnifying eyepiece.

PICTURING SPACE

Astronomers learn about space using telescopes. These telescopes are designed to each pick up one particular type of electromagnetic radiation from space, and use that radiation to create an image. The pictures on the right here show the Crab Nebula viewed through different types of telescopes.

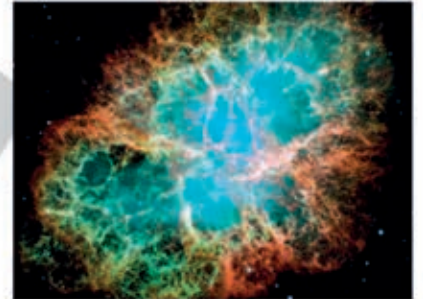
INFRARED TELESCOPE

These detect heat given off by objects. They are often used in space, where they are kept cold and far from Earth (so that they do not pick up confusing heat data from objects on Earth).



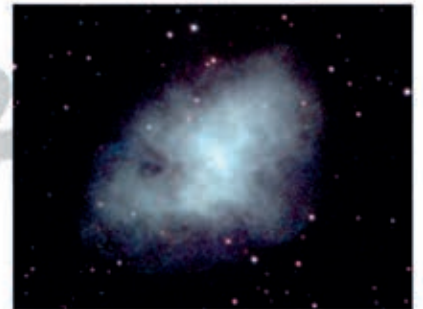
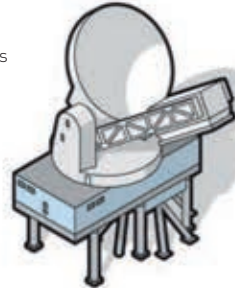
OPTICAL TELESCOPE

These use lenses and mirrors to capture light from distant objects. Reflecting and refracting telescopes are forms of optical telescopes.



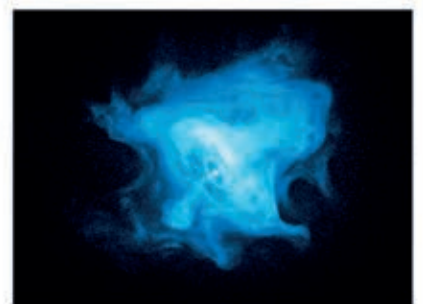
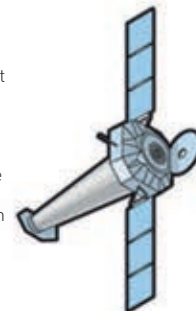
ULTRAVIOLET TELESCOPE

Hot and active objects in the cosmos give off large amounts of ultraviolet energy, so they are revealed in most detail with this kind of telescope.



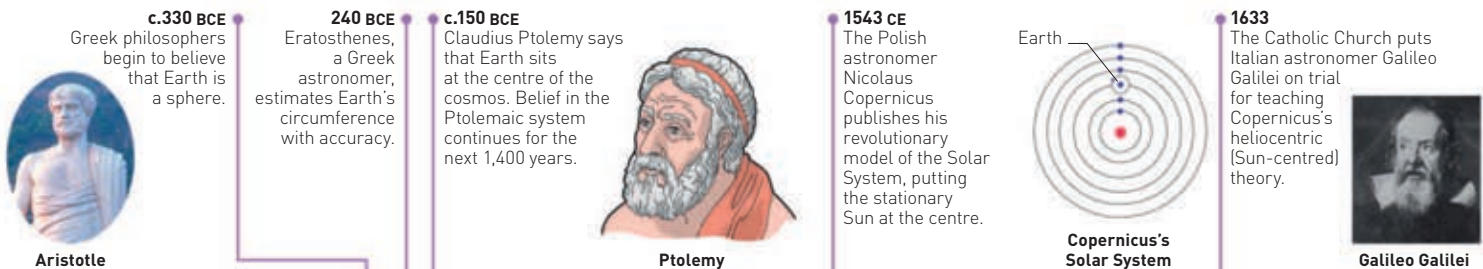
X-RAY TELESCOPE

These telescopes capture high-energy rays from extremely hot objects. X-rays from celestial objects are partly blocked by the Earth's atmosphere, so these telescopes are sent into space. They gather information from space objects as they orbit around Earth.



TIMELINE

Since ancient times, people have recorded astronomical observations. As science advances, we are still trying to discover the great mysteries of the Universe.



2500 BCE

2500 BCE
Building of Stonehenge. The stones here mark the rising and setting points of the Sun at the solstices.



Stonehenge

700 BCE
Babylonians predict regular patterns of Sun and Moon eclipses.



Aristarchus's calculations

280 BCE
Ancient Greek astronomer Aristarchus calculates the size of the Sun and Moon and their distances from Earth.

240 BCE
The first certain appearance of Halley's Comet is described in the Chinese *Records of the Grand Historian*.



Halley's Comet

1054 CE
Chinese astronomers observe a supernova that is visible in the daytime. The matter blasted outwards by it remains observable as the Crab Nebula.



Elliptical orbit

1609
German mathematician Johannes Kepler calculates that the planets follow noncircular, elliptical orbits.



Isaac Newton

1687
English scientist Isaac Newton discovers that gravity keeps the Moon in orbit around Earth, and the planets in orbit around the Sun.